

Husayn El Sharif

Senior Data Scientist • Machine Learning Engineer

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Professional Summary

Data Scientist and ML Engineer with 10+ years of experience developing data-driven solutions across environmental, engineering, and business domains. Ph.D. in Civil Engineering from Georgia Tech with expertise in machine learning, Generative AI, RAG, deep learning, geospatial analytics, and cloud platforms. Google Cloud Professional ML Engineer and Databricks Certified Generative AI Engineer Associate.

Technical Skills

Programming: Python, SQL, R, MATLAB

Machine Learning: Scikit-Learn, TensorFlow, PyTorch, XGBoost, Deep Learning

Data Engineering & MLOps: NumPy, Pandas, PySpark, MLflow, Airflow, Streamlit, Docker, Git

AI Engineering: Prompt Engineering, Fine-Tuning, Agentic RAG, Claude Code, Codex

Cloud & Data Platforms: Google Cloud, AWS, Databricks, Snowflake, Hugging Face

Visualization: Matplotlib, Seaborn, Plotly, Tableau, Power BI

Geospatial Analysis: ArcGIS, QGIS, GeoPandas, Google Earth Engine

Certifications

- Google Cloud Professional Machine Learning Engineer (2026)
- Databricks Certified Generative AI Engineer Associate (2026)

Selected Machine Learning Projects

Remote Sensing & ML for Water Quality Assessment at Lake Lanier

Built a satellite-based ML pipeline with Sentinel-2 and Google Earth Engine to estimate chlorophyll-a concentrations and detect harmful algal blooms, enabling near-real-time water quality monitoring for Gwinnett County water operations.

[View Project](#)

Smart Meter Risk Segmentation with Unsupervised ML (AMI Data)

Built a k-means clustering pipeline on high-frequency AMI data to segment smart meters by load and power-quality behavior, enabling identification of high-risk assets and supporting proactive maintenance prioritization for utility operations. [View Project](#)

AI-Powered Retinal Disease Detection with Deep Learning & Computer Vision

Developed a deep learning computer vision pipeline for retinal disease classification, achieving 89% accuracy and a 0.88 macro F1 score across four diagnostic classes, supporting automated screening and clinical decision support.

[View Project](#)

Agentic RAG Workflow for Building & Fire Code Review

Built a LangChain agentic RAG app with hybrid keyword-semantic search for Cobb County building and fire code Q&A across ordinances, permit forms, and code references; improved faithfulness from 0.26 to 0.97, achieving 0.77 context precision and 0.72 context recall. [View Project](#)

Professional Experience

Senior Data Scientist (Part-Time)

Sept 2025 – present

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Marietta, GA

- Developed a property lead-scoring pipeline using weather, permit, valuation, and geospatial data; outperformed a heuristic baseline with logistic regression and XGBoost, identifying a projected 12% revenue lift (~\$450K per 600 leads) and 3.3x higher PR-AUC. Added Platt scaling and SHAP for explainable, per-lead insights.
- Built an explainable RAG platform to match project requirements with qualified engineering firms, indexing 7K+ public profiles into ~70K retrieval documents using PostgreSQL/pgvector, embeddings, hybrid search, reciprocal rank fusion, and reranking with a cross encoder.
- Built an AI-powered lead-enrichment pipeline with Python, Google Gemini, and the Google Places API to surface high-value industrial prospects for an engineering consultancy, replacing manual prospecting with a scored outreach list and accelerating lead qualification.

Senior Data Scientist (Research Engineer)

Aug 2012 – present

Georgia Institute of Technology, School of Civil & Environmental Engineering

Atlanta, GA

- Built Dockerized environments for geospatial ML workflows, supporting reproducible deployment of satellite-based environmental monitoring pipelines.
- Developed ML models integrating soil, crop, and climate data to predict crop yield and irrigation demand, enabling data-driven water resource planning and quantifying long range risks to agricultural production in the Apalachicola-Chattahoochee-Flint River Basin.
- Oversaw \$400K+ in annual federal and state research funding as Assistant Director of the Georgia Water Resources Institute.
- Mentored graduate researchers in data science and ML methods, supporting publications and conference presentations.

Engineering Consultant (Part-Time)

Aug 2022 – Aug 2025

4Earth, Inc.

Kennesaw, GA

- Developed physics- and data-driven system models and AI-Ops pipelines for real-time AWS IoT sensor analytics, improving engineering efficiency and reducing delivery time by 20%.

Data Scientist (Graduate Research Assistant)

May 2010 – Aug 2012

Florida Atlantic University, Department of Civil, Environmental, & Geomatics Engineering

Boca Raton, FL

- Built and validated ML models to reconstruct missing environmental data, improving accuracy and spatial consistency for hydrologic and climate analyses.

Education

Ph.D.	Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2019
M.Sc.	Civil Engineering	Georgia Institute of Technology	Atlanta, GA	GPA: 3.97	May 2016
B.Sc.	Civil Engineering	Florida Atlantic University	Boca Raton, FL	GPA: 3.81	Aug 2011